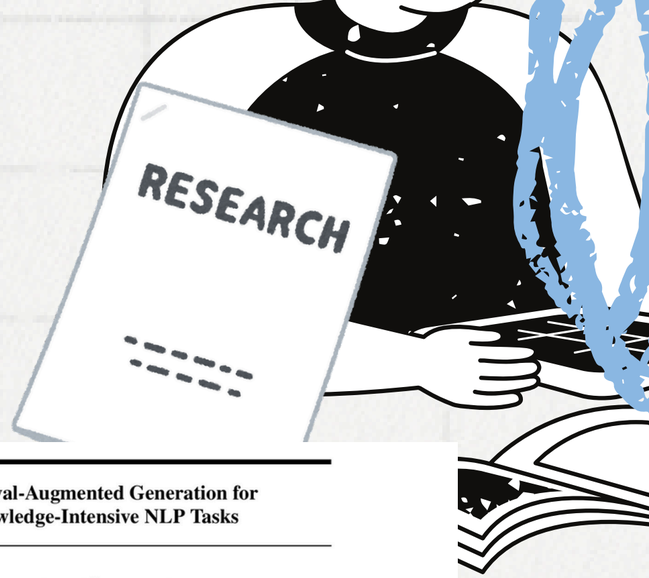
Doctor Pre-Visit Guidance Generator using LLM with RAG

1905074 - Tahmid Islam Tomal
1905085 - Md Hasibul Hasan

Group: B1_7

Details and Papers



This model combines an LLM for general medical guidance with a RAG (Retrieval-Augmented Generation) layer for personalized, specific instructions. The LLM provides standard information, while RAG retrieves patient-specific documents or guidelines. Together, they generate tailored, accurate care instructions.

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis[†‡], Ethan Perez[⋆], Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†], Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel[†‡], Sebastian Riedel[†‡], Douwe Kiela[†]

[†]Facebook AI Research; [‡]University College London; [⋆]New York University; plewis@fb.com

https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html

Benchmarking Large Language Models in Retrieval-Augmented Generation

Jiawei Chen[1,3], Hongyu Lin[1,*], Xianpei Han[1,2,*], Le Sun[1,2]

[1]Chinese Information Processing Laboratory, Institute of Software, Chinese Academy of Sciences, Beijing, China
[2]State Key Laboratory of Computer Science, Institute of Software, Chinese Academy of Sciences, Beijing, China
[3]University of Chinese Academy of Sciences, Beijing, China
{jiawei2020,hongyu,xianpei,sunle}@iscas.ac.cn

https://doi.org/10.1609/aaai.v38i16.29728

Dataset

Training dataset link (doctor_qa_bangla)

https://huggingface.co/datasets/shetumohanto/doctor_qa_bangla/viewer/default/train

Size of downloaded dataset files:
5.66 MB

Size of the auto-converted Parquet files:
1.99 MB

Number of rows:
5,135

```
import pandas as pd

df = pd.DataFrame(dataset)
df.info()
```

[5] ✓ 0.7s

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5135 entries, 0 to 5134
Data columns (total 1 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   train   5135 non-null   object
dtypes: object(1)
memory usage: 40.2+ KB
```

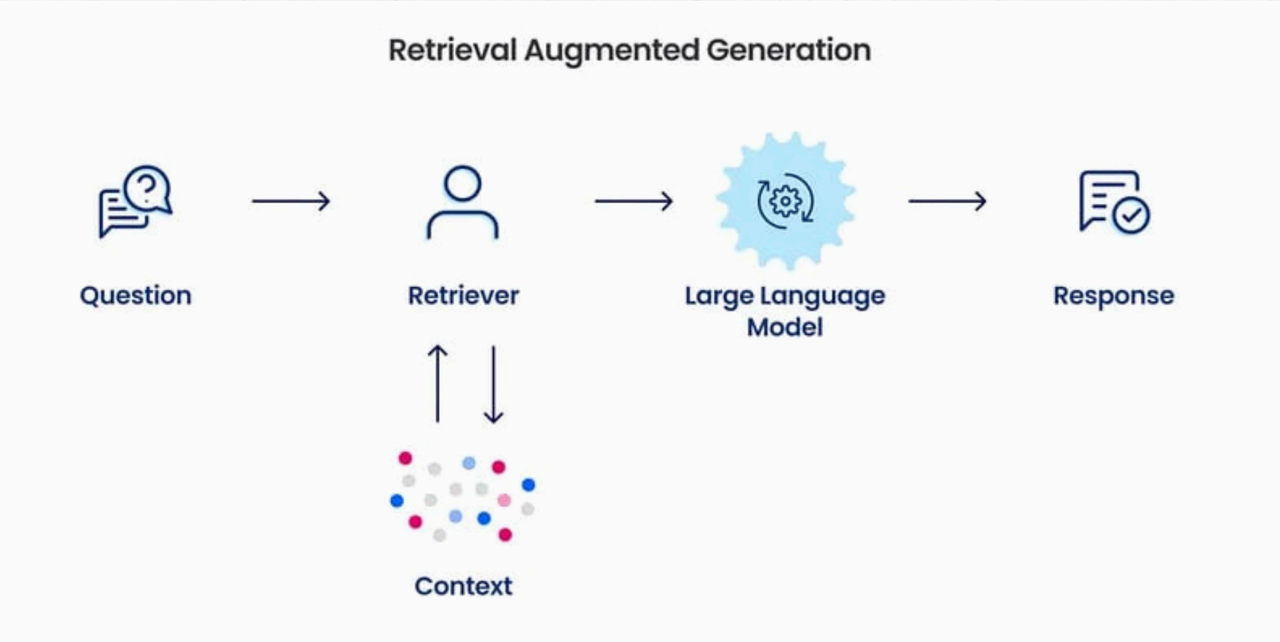
```
df.head(5)
```

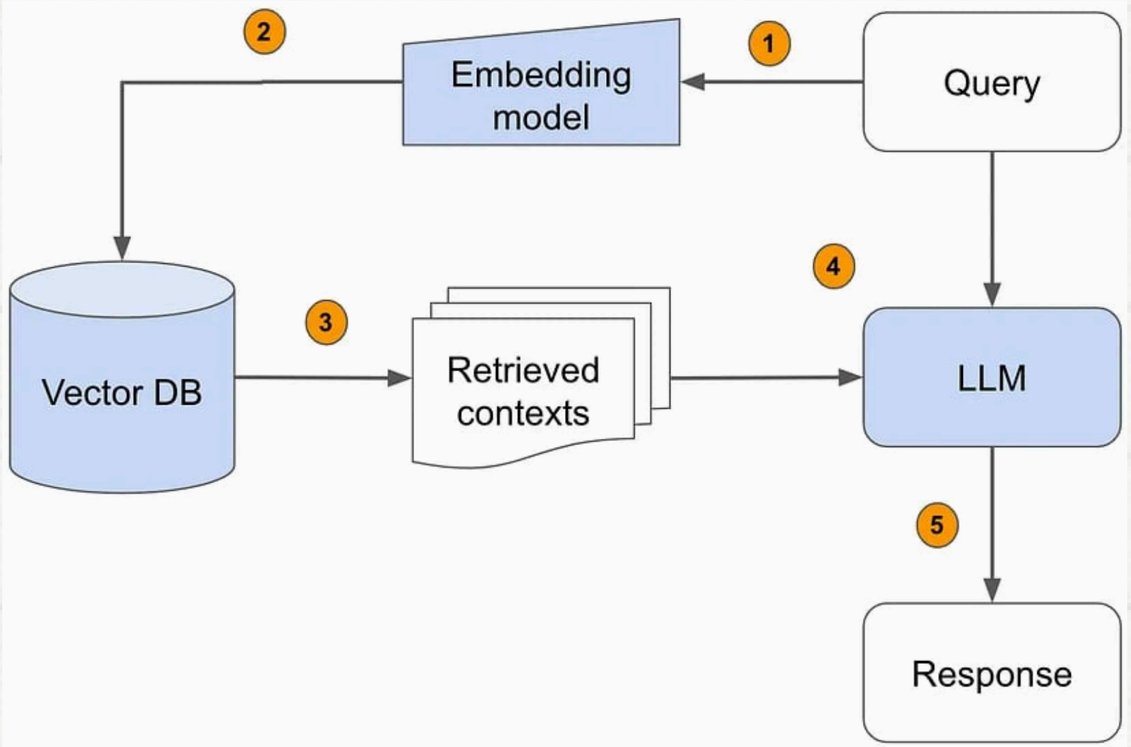
✓ 0.0s

```
0    {'text': '<s> [INST] আমার ৩ বছর এর
1    {'text': '<s> [INST] আমার মেয়ের বয়স ১
2    {'text': '<s> [INST] আমার বাচ্চার বয়স
3    {'text': '<s> [INST] আমার বাচ্চার বয়স
4    {'text': '<s> [INST] আমার মেয়ের বয়স ৬
```



Architecture





Performance Metrics

1 Precision@k

Assesses the relevance of documents in the top k retrieved results, ensuring that retrieved information is highly relevant to the patient query.

2 Response Time

Tracks the latency between query input and response, which is crucial for usability in real-world settings.

3 Feedback Incorporation Rate:

Tracks how well the model adapts based on flagged or corrected outputs.

RAG Performance Metrics: The Future of LLM Evaluation



Boris Villazon-Terrazas, PhD
Global Gen AI CoE Leader | Europe West AI Innovation Leader | AI & DS Product Manager | CAIO | CTO | Mentor | People...

June 23, 2024

RAGAS: A New Benchmark for QA Systems

RAGAS stands out as an innovative framework designed to evaluate QA pipelines in novel ways. It provides a comprehensive set of metrics that scrutinize both the retriever and generator components of a RAG system. By measuring aspects such as answer correctness, faithfulness, context relevancy, and precision, RAGAS offers a granular view of a system's performance [1].

TrueLens: Seeing Through the Lens of Accuracy

While RAGAS focuses on the evaluation process, TrueLens contributes by enhancing the accuracy of these assessments. It's an approach that complements the RAG Triad of metrics, providing deeper insights into the effectiveness of RAG applications [2]